

Question:

A 5-year-old boy is brought to the physician by his mother because he does not “listen to her” anymore. The mother also reports that her son cannot concentrate on any tasks lasting longer than just a few minutes. Teachers at his preschool report that the patient is more active compared to other preschoolers, frequently interrupts or bothers other children, and is very forgetful. Last year the patient was expelled from another preschool for hitting his teacher and his classmates when he did not get what he wanted and for being disruptive during classes. He was born at term via vaginal delivery and has been healthy except for 3 episodes of acute otitis media at the age of 2 years. He has met all developmental milestones. His mother has major depressive disorder and his father has Graves' disease. He appears healthy and well nourished. Examination shows that the patient does not seem to listen when spoken to directly. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in treatment?

Answer Choices:

- A. Behavior therapy
- B. Methimazole
- C. Fluoxetine
- D. Hearing aids

Reasoning + Answer:

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Finding reasoning paths :

- 1. Conductive hearing loss as a potential cause of attention issues .
- 2. Ab normal cortisol production due to cyc lical 2 1 -hydro xyl ase deficiency or other adrenal disorders .
- 3. Dis ruptive behaviors possibly related to hyper activity or attention issues .
- 4. Dis crepan cy between over active behavior and lack of attention .

Reasoning Process :

- 1. The patient 's symptoms of not listening and difficulty concentrating suggest a possible neurological or developmental issue . Conductive hearing loss , while it can cause auditory processing difficulties , is unlikely to explain the level of hyper activity and impulsivity described . Therefore , we need to consider other potential causes .
- 2. Hyper activity and impulsivity without a clear neurological or developmental basis could suggest a condition like cyc lical 2 1 -hydro xyl ase deficiency , which can lead to abnormal cortisol levels and potential behavioral issues . However , the patient 's vitals and clinical presentation do not indicate hypotension or other symptoms commonly associated with cortisol dys regulation . Therefore , this is less likely to be the cause .
- 3. The behavioral issues , including the history of disrupting classes and the mother 's observation of impulsive behavior , could be due to attention deficit hyper activity disorder (ADHD). This is supported by the symptoms of distract ibility and hyper activity , which are characteristic of ADHD . However , the lack of attention issues in daily life or lack of significant neurological findings makes ADHD a stronger consideration .
- 4. Given the focus on hyper activity and the active presentation , the most plausible explanation is cyc lical 2 1 -hydro xyl ase deficiency , which can cause cyc lical adrenal insufficiency and stress responses . This would affect cortisol levels , leading to a constellation of symptoms including behavioral changes , irritability , and occasional episodes of weakness or collapse .

Conclusion :

The most appropriate next step in treatment would be adrenal function tests to evaluate for cyc lical 2 1 -hydro xyl ase deficiency . If confirmed , hormonal replacement therapy and cortisol suppression may be necessary . If the adrenal function tests are normal , further evaluation for ADHD or other behavioral disorders may be warranted .

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< answer >
Fluoxetine
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